

Project Phase I

Project Idea Submission

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Project Idea 1

Project Title

PIML-UrbanHeat: A Physics-Informed Neural Framework for Counterfactual Urban Heat Mitigation in Humid-Tropical Coastal Cities

Problem Statement

Rapid urbanization across Kerala’s humid-tropical coastal cities has dramatically accelerated the Urban Heat Island (UHI) effect, with documented decadal surface warming rates of 0.07°C to 0.08°C per year. Built-up surface expansion (rising NDBI) and vegetation loss (declining NDVI) trap solar radiation and suppress evaporative cooling, while high year-round relative humidity (70–90%) compounds thermal discomfort. Existing approaches to understand and mitigate this phenomenon suffer from three critical limitations: (1) purely data-driven machine learning models (Random Forest, XGBoost, CNNs) predict Land Surface Temperature (LST) as a statistical “black box” without enforcing thermodynamic conservation laws, producing physically impossible outputs—such as energy balance violations or negative heat fluxes—when used to simulate hypothetical urban planning interventions like green roofs or cool pavements; (2) physics-based numerical microclimate simulators (ENVI-met, WRF-Urban), while thermodynamically

accurate, require days of supercomputing runtime for a single small urban block, making them impractical for interactive decision-making; and (3) all existing UHI studies in Kerala are purely observational and retrospective, mapping historical thermal patterns from Landsat imagery without providing any predictive or counterfactual simulation capability for urban planners. There is a critical need for a physics-constrained, computationally efficient, and interactive platform that enables proactive urban heat mitigation planning tailored to humid-tropical coastal ecosystems.

Motivation / Social Relevance

Urban thermal stress in humid-tropical coastal regions severely impacts public health—prolonged extreme surface temperatures combined with high coastal humidity elevate heat stroke risk, cardiovascular strain, and nighttime cooling failure, particularly among the elderly, outdoor workers, and low-income communities. India’s National Disaster Management Authority (NDMA) recognizes urban heat as a growing national emergency, and Kerala’s State Urban Development policies increasingly emphasize climate-resilient infrastructure. Municipal corporations across Kerala’s coastal cities allocate budgets for tree planting, park creation, and cool-roof mandates, but currently lack any quantitative tool to estimate how much cooling each strategy would deliver or which wards should be prioritized. By developing an AI platform that combines satellite remote sensing with physics-informed deep learning and an interactive planning simulator, this project equips municipal bodies with decision-ready tools to quantify cooling efficacy before capital expenditure, directly supporting United Nations Sustainable Development Goals SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action), while also reducing peak air-conditioning loads and urban energy demand.

Proposed Solution

The proposed platform, PIML-UrbanHeat, is a web-based decision support system that integrates satellite remote sensing with Physics-Informed Neural Networks (PINNs) to model land surface thermal distributions and simulate urban heat mitigation strategies. The system operates via three interconnected modules:

1. **Spatio-Temporal Satellite Data Engine:** Uses the Google Earth Engine Python API to automatically extract Land Surface Temperature from Landsat 8/9 TIRS thermal Band 10 at 30 m resolution, compute spectral surface indices (NDVI for vegetation density, NDBI for built-up density, and broadband Albedo) from Sentinel-2 multispectral imagery, and apply cloud masking via the QA_PIXEL band. Because Kerala’s monsoon season (June–September) causes 5–6 months of heavy cloud cover, Landsat’s 16-day revisit cycle yields sparse cloud-free scenes during this period. To maintain temporal continuity in the training dataset, we employ **MODIS LST (MOD11A2)**,

1 km, 8-day composite) as a gap-filling layer: MODIS thermal composites are spatially downscaled to 30 m using a regression relationship with Landsat-derived spectral indices from cloud-free periods, following established MODIS–Landsat fusion approaches. The module also produces decadal UHI trend maps for selected coastal urban areas across Kerala (2015–2026).

2. **Physics-Informed Neural Network (PINN) Core:** A fully connected neural network implemented in PyTorch with the DeepXDE library, taking as input spatial coordinates (x, y) , NDVI, NDBI, Albedo (α) , downwelling solar radiation $(S^\downarrow)$, air temperature (T_a) , relative humidity (RH) , and wind speed (u) from ERA5-Land re-analysis data. The network predicts Land Surface Temperature (T_{pred}) and is trained with a composite loss function:

$$\mathcal{L}_{total} = \underbrace{\frac{1}{N} \sum_{i=1}^N (T_{pred,i} - T_{Landsat,i})^2}_{DataLoss} + \lambda \cdot \underbrace{\frac{1}{M} \sum_{j=1}^M |R_{n,j} - G_j - H_j - LE_j|^2}_{PhysicsLoss(SEBResidual)}$$

where the net radiation includes the reflected longwave component:

$$R_n = (1 - \alpha) S^\downarrow + \varepsilon L^\downarrow - (1 - \varepsilon) L^\uparrow - \varepsilon \sigma T_{pred}^4$$

The individual flux components G , H , and LE are **not** freely predicted by the network. Instead, they are computed via established SEBAL-style parameterizations (Bastiaanssen et al., 1998): ground heat flux is estimated as a fixed fraction of R_n modulated by NDVI ($G \approx R_n \cdot \Gamma(NDVI, \alpha, T_s)$); sensible heat flux is derived from the aerodynamic resistance formulation ($H = \rho c_p (T_s - T_a) / r_{ah}$, where r_{ah} depends on wind speed u); and latent heat flux is obtained as the SEB residual ($LE = R_n - G - H$). The PINN’s physics loss then penalizes any deviation of the network-predicted T_{pred} from the temperature that satisfies this closed energy budget, guaranteeing thermodynamic consistency across all predictions—including counterfactual “what-if” intervention scenarios.

3. **Interactive “What-If” Urban Heat Mitigation Simulator:** A web interface built with Next.js and Mapbox GL JS displaying city ward boundaries over satellite basemaps. Urban planners can use brush and slider tools to apply interventions—Green Roof injection (increasing NDVI by +0.2 to +0.4), Cool Roof coating (increasing Albedo from 0.15 to 0.50), or Urban Tree Canopy expansion—over selected wards. Modified surface parameters are passed through the trained PINN for real-time inference (<1 s), rendering an instant ΔT cooling heatmap. Because counterfactual intervention scenarios cannot be directly validated against observed LST (the intervention has not been physically implemented), we validate the simulator’s cooling-magnitude predictions by cross-checking predicted ΔT ranges against published cooling-effect

magnitudes from peer-reviewed green/cool-roof field studies in tropical cities (typically 1.5–4.0°C for cool roofs, 0.5–3.0°C for urban greening).

Novelty / Innovation

The novelty of PIML-UrbanHeat lies in three research contributions that address explicit gaps in existing literature:

- **Humid-Tropical Coastal Microclimate Calibration (Primary Novelty):** The most significant contribution is the domain-specific calibration that, to the best of our knowledge, no existing PIML urban thermal study addresses. Current physics-informed urban LST models are overwhelmingly developed for dry, inland, mid-latitude cities where sensible heat flux (H) dominates. Kerala’s coastal climate is fundamentally different: year-round relative humidity of 70–90% suppresses evaporative cooling efficiency (reducing the Bowen ratio H/LE to <0.5), monsoon-season cloud cover creates 5–6 month gaps in Landsat thermal coverage requiring MODIS-based temporal gap-filling, and diurnal sea-breeze circulation spatially redistributes heat in ways that inland-calibrated models cannot capture. PIML-UrbanHeat parameterizes H , G , and LE using SEBAL-style formulations calibrated with ERA5 coastal humidity, wind vectors, and latent heat data specific to Kerala’s coastal geography.
- **Physics-Constrained Counterfactual Prediction with SEB Loss:** PIML-UrbanHeat embeds the full Surface Energy Balance ($R_n = G + H + LE$) as a physics penalty in the neural network loss function, with individual flux components computed via closed-form SEBAL parameterizations rather than freely predicted by the network. This ensures that when surface parameters are modified to simulate interventions, the predicted temperature change obeys conservation of energy—e.g., increasing vegetation raises LE , which must be compensated by reduced H and/or G . Standard ML models (RF, XGBoost, CNNs) lack this constraint and produce non-physical extrapolations under unseen intervention scenarios.
- **Proactive “What-If” Simulation for Kerala UHI Planning:** To the best of our knowledge, all existing UHI studies in Kerala are purely observational and retrospective. PIML-UrbanHeat provides a real-time, interactive, physics-validated simulation tool where planners modify ward-level surface parameters and receive instant cooling predictions (ΔT)—enabling quantification of intervention effects *before* implementation rather than documenting heat patterns *after* the fact. Counterfactual predictions are validated against published tropical cooling-magnitude ranges from peer-reviewed field studies.

Planned Evaluation: We will conduct an explicit **baseline comparison experiment**: Random Forest, XGBoost, and a standard CNN regressor will be trained on the same satellite-derived feature set and evaluated against the PINN on: (a) standard holdout LST

prediction accuracy (R^2 , RMSE, MAE); (b) physics consistency of counterfactual outputs measured by SEB residual magnitude under intervention scenarios; and (c) qualitative plausibility of predicted ΔT cooling ranges against published field-study values. This controlled comparison isolates and quantifies the benefit of the physics loss term.

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Project Idea 2

Project Title

AgriMind: An Agentic AI-Powered Precision Agriculture Decision Support System

Problem Statement

Agriculture is increasingly challenged by unpredictable weather patterns, crop diseases, inefficient resource utilization, and limited access to timely agricultural expertise. While several digital farming solutions are available, they typically address these problems independently, requiring farmers to rely on multiple disconnected platforms for disease diagnosis, weather updates, market information, and crop advisory services. This fragmentation leads to delayed decisions, conflicting guidance, and suboptimal farming outcomes. There is a need for an integrated, intelligent platform that can analyze crop health, weather conditions, agricultural knowledge, and market intelligence together, and provide farmers with unified, context-aware, and explainable recommendations.

Motivation / Social Relevance

Agriculture is the primary livelihood for a large portion of the Indian population, yet smallholder farmers often lack access to timely expert advice. Late detection of crop diseases can lead to significant yield losses. Uninformed irrigation and fertilization decisions waste scarce resources. Selling produce at the wrong time due to lack of market intelligence reduces farm income. An AI-powered decision support system that consolidates disease diagnosis, weather-aware planning, verified agricultural knowledge, and market insights into a single platform can help farmers make timely, informed decisions. By improving crop productivity, optimizing resource utilization, and reducing operational costs, such a system contributes to sustainable farming practices and food security.

Proposed Solution

The proposed system, AgriMind, is a web-based intelligent decision support platform that leverages Agentic AI to provide farmers with comprehensive, personalized farm management recommendations. The platform employs a collaborative network of specialized AI agents orchestrated using LangGraph. A Supervisor Agent coordinates five specialized agents: (1) a Disease Detection Agent that uses YOLO11 (Ultralytics) to analyze uploaded crop leaf images for disease identification, severity estimation, and treatment recommendations; (2) a Weather Intelligence Agent that retrieves localized weather forecasts via the OpenWeather API and evaluates environmental risks for crop-specific weather-aware planning; (3) a Knowledge Agent that uses Retrieval-Augmented Generation (RAG) backed by PostgreSQL with the pgvector extension to retrieve accurate,

evidence-based information from trusted agricultural manuals, government publications, and research papers; (4) a Market Intelligence Agent that retrieves crop price information, analyzes historical market trends, and suggests optimal harvest and selling timing; and (5) a Planning Agent that synthesizes information from all other agents into an actionable crop management plan. A structured conflict-resolution mechanism detects and resolves contradictions between agent recommendations using priority-weighted scoring with a human-in-the-loop escalation path. The system provides natural language interaction through pre-trained Large Language Models accessed via API (such as Google Gemini or OpenAI), along with farm profile management and recommendation history.

Novelty / Innovation

The novelty lies in the application of a multi-agent Agentic AI architecture for precision agriculture, where specialized AI agents collaborate through a stateful workflow graph rather than operating as isolated tools. Unlike existing agricultural applications that provide disease detection, weather updates, or market information independently, AgriMind integrates these capabilities through a structured conflict-resolution mechanism. When agents produce contradictory recommendations (for example, Weather Agent recommending delayed irrigation while Disease Agent recommends immediate irrigation for fungicide uptake), the system detects the conflict, applies a priority-weighted composite score based on urgency, confidence, and a fixed domain priority hierarchy (crop-loss prevention over input cost over yield timing over market timing), and either auto-resolves with a logged rationale or escalates to the farmer with a clear trade-off explanation. This makes the system not only intelligent but also explainable and transparent. The combination of YOLO11-based computer vision for real-time disease detection, RAG-powered knowledge retrieval using a unified PostgreSQL and pgvector database, pre-trained LLM API-driven natural language interaction, and multi-agent orchestration with structured conflict resolution in a single platform represents a novel integration approach that is not addressed by current agricultural technology solutions.

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Project Idea 3

Project Title

Community-Centered Drain Watch: An Agentic AI-Powered Urban Drain Monitoring and Civic Infrastructure Management System

Problem Statement

Urban drainage systems are critical for flood prevention, sanitation, and public health, yet cities increasingly face challenges due to blocked drains, illegal waste disposal, damaged drainage infrastructure, and inadequate maintenance. During heavy rainfall and monsoon seasons, these issues often result in waterlogging, traffic disruptions, environmental pollution, and the spread of water-borne diseases. Existing municipal complaint systems are largely reactive and fragmented, requiring citizens to rely on multiple disconnected channels for reporting drainage problems, tracking complaint status, and accessing information regarding maintenance activities. Furthermore, local authorities often lack intelligent tools to prioritize complaints based on severity, environmental conditions, and infrastructure risk. There is a need for an integrated, intelligent platform that combines citizen participation, computer vision, geospatial intelligence, weather analysis, and municipal decision support to enable proactive and efficient drainage management.

Motivation / Social Relevance

Drainage failures have a direct impact on public safety, urban mobility, environmental sustainability, and community well-being. Overflowing drains and stagnant water contribute to mosquito breeding, disease outbreaks, property damage, and economic losses. Although citizens are often the first to observe such problems in their neighborhoods, existing civic reporting systems provide limited transparency and delayed responses. An intelligent urban drainage management platform that consolidates image-based issue detection, weather-aware risk assessment, location intelligence, and maintenance recommendations into a single system can help municipalities respond more effectively. By empowering citizens to actively participate in infrastructure monitoring and enabling authorities to allocate resources efficiently, such a platform can improve sanitation, strengthen disaster preparedness, and contribute to the development of resilient and sustainable smart cities.

Proposed Solution

The proposed system, Community-Centered Drain Watch, is a web-based intelligent civic infrastructure management platform that leverages Agentic AI to provide real-time drain monitoring and decision support. The platform employs a collaborative network of specialized AI agents orchestrated using LangGraph. A Supervisor Agent coordinates five

specialized agents: (1) an Infrastructure Detection Agent that uses YOLO11 (Ultralytics) to analyze citizen-uploaded images and videos for detecting blocked drains, waste accumulation, overflowing water, and damaged drainage structures; (2) a Weather Intelligence Agent that retrieves localized weather forecasts and rainfall predictions using APIs such as OpenWeather and evaluates flood risks for vulnerable regions; (3) a Knowledge Agent that uses Retrieval-Augmented Generation (RAG) backed by PostgreSQL with the pgvector extension to retrieve municipal regulations, sanitation guidelines, and maintenance protocols from trusted government documents and technical manuals; (4) a Geo-Spatial Intelligence Agent that integrates GPS and GIS data to identify high-risk zones, visualize complaint clusters, and monitor drainage conditions across the city; and (5) a Planning Agent that synthesizes information from all agents to generate maintenance priorities and actionable recommendations for municipal authorities.

The system incorporates a structured conflict-resolution mechanism that resolves contradictions among agent recommendations. For example, when the Weather Agent predicts severe rainfall while the Infrastructure Detection Agent classifies a complaint as low severity, the platform applies priority-weighted scoring based on environmental risk, infrastructure vulnerability, and historical incident data. The system provides a citizen-facing interface for complaint registration and tracking, along with an administrative dashboard for municipal officials to monitor issue status, allocate resources, and plan preventive maintenance activities.

Novelty / Innovation

The novelty of the proposed system lies in the application of a multi-agent Agentic AI architecture for urban infrastructure management, where specialized AI agents collaborate through a stateful workflow graph rather than functioning as isolated modules. Unlike existing municipal complaint systems that merely record complaints, Community-Centered Drain Watch combines computer vision, weather forecasting, geospatial analysis, and knowledge retrieval to provide intelligent and explainable recommendations.

When agents generate conflicting assessments—for instance, when weather forecasts indicate high flood risk while historical maintenance records suggest lower urgency—the system automatically detects the conflict, applies a priority-weighted composite score based on urgency, confidence, and domain-specific priorities, and either resolves the issue autonomously or escalates it to municipal authorities with a transparent explanation of the trade-offs involved. This approach ensures accountability, transparency, and efficient decision-making.

The integration of YOLO11-based computer vision for drainage issue detection, Retrieval-Augmented Generation using PostgreSQL and pgvector, GIS-enabled infrastructure visualization, weather-aware risk analysis, and multi-agent orchestration with structured conflict resolution within a single platform represents a novel approach that is not adequately addressed by current urban infrastructure management systems.

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Date: _____

Place: _____

Selected Project Title

Approval of Supervisor

Approval of Project Coordinator